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Multivariate Analysis

Dataset: Airbnb Listings

Explaining Airbnb Price Differences: A Comparative Analysis of Boston and Jersey City

Introduction

Airbnb and other short-term rental platforms have transformed urban housing markets by providing flexible accommodations for travelers while also creating new opportunities for property owners. The pricing of Airbnb listings is influenced by a wide range of factors, including the type of accommodation, the number of guests it can accommodate, the location, neighborhood characteristics, and the amenities offered. Amenities such as private bathrooms, kitchens, Wi-Fi, or premium services often make listings more attractive to guests and are associated with higher prices.

Understanding how these features affect pricing is important for guests and hosts. For hosts, it informs decisions about which amenities to offer and how to price their properties competitively. For travelers, on the other hand, it provides insight into how accommodation choices and location affect costs. For researchers and policymakers, examining the differences between listing characteristics and pricing can shed light on trends in urban housing markets and tourism demand.

Comparisons across different cities, such as Jersey City and Boston, show how local market conditions, neighborhood dynamics, and supply and demand patterns can contribute to pricing variability. This highlights not only the economic factors that influence short-term rental markets but also the ways in which urban environments shape the experiences and expectations of travelers.

This topic is particularly relevant to the Cornell student body, as many students travel to cities such as Jersey City and Boston during academic breaks for leisure, internships, or networking opportunities. Short-term rentals often serve as a primary housing option during these visits, making cost an important consideration. By examining how Airbnb prices vary between these two cities and what factors contribute to those differences, this study can provide useful insight into whether one city may be more affordable than the other for temporary housing. This perspective highlights the practical value of understanding pricing dynamics, as it can help inform more cost-effective travel and housing decisions for students.

Research Question

The main objective of this study is to examine how Airbnb prices per guest differ between Jersey City and Boston, and to what extent these differences can be explained by listing characteristics such as specific amenities (e.g., the presence of a patio or balcony), neighborhood location, and the number of guests each listing can accommodate. Price is treated as the primary outcome variable because it reflects both the listing's value and market conditions in each city. To make comparisons more meaningful, we adjust listing prices to a per-guest basis. This approach accounts for differences in listing size and allows for a fairer comparison of value across properties of varying capacities.

In addition to location, several key factors may influence pricing. Amenities play a significant role, as listings that offer more or higher-quality features, such as kitchens, private bathrooms, or premium services, may command higher prices. Neighborhood characteristics further contribute to price variation by capturing differences in accessibility, desirability, and proximity to major attractions.
To ensure meaningful comparisons across cities, the analysis includes data cleaning steps such as adjusting and standardizing prices, as well as harmonizing amenity descriptions to account for variation in how hosts report features. The dataset is also filtered by the number of guests accommodated to maintain consistency across listings being compared.

By considering these factors jointly, the study aims to isolate how amenities and other listing characteristics influence Airbnb prices and to determine whether systematic pricing differences exist between Jersey City and Boston. This approach provides insight into how temporary housing markets function across different urban environments and how specific features contribute to variation in listing prices.

In this study, we focus on a selected set of c relevant amenities for college students to better understand how specific features contribute to Airbnb pricing. Rather than analyzing every possible amenity, we focused on a subset that reflects both everyday convenience and potential "premium" value to guests. These include household and comfort features such as heating, air conditioning, hot water, and cooking basics, as well as lifestyle and convenience amenities like parking, gym/workout equipment, patio or balcony access, and pet allowances. Additionally, we include widely expected baseline amenities such as television and dishes/silverware to capture differences between standard and more "premium" listings. By narrowing our focus to these amenities, we are able to more clearly isolate their individual and combined effects on price per guest, while also ensuring consistency across listings in both Jersey City and Boston. This approach allows for a more interpretable comparison of how specific features contribute to pricing differences between the two cities.

Main Variables:

Table 1: Variables and Corresponding Names

Variable Name	**Information**
Neighborhood_cleansed	The neighborhood that the listing resides in
Accommodates	Number of people that can be housed in the listing
Amenities	A list of functional Airbnb amenities offered
Price	The cost of the Airbnb per night
Price_with_tax(New variable)	The cost of the Airbnb per night after adjusting for tax
Price_per_guest(New variable)	The rate charged per individual guest at the Airbnb after adjusting for tax

Challenges We Faced

One of the primary challenges in this study was the lack of consistent and complete data across cities. Our original goal was to compare Airbnb listings in New York City and Boston to analyze how amenities influence pricing. However, we found that the New York City dataset did not include price information, which is essential as the dependent variable in our analysis. To address this issue, we explored several alternative approaches to obtain pricing data. First, we considered scraping prices directly from the Airbnb website, but this proved impractical due to the large volume of listings and the time required for manual data collection. Additionally, Airbnb's platform includes technical and policy-based restrictions such as dynamic content, rate limiting, and terms of service that make large-scale data extraction difficult. We also attempted to use listing URLs from the dataset to retrieve pricing information, but many links were inactive, and among those that remained accessible, pricing data was inconsistent or difficult to extract. Finally, we explored publicly available datasets on platforms like Kaggle; however, these datasets often lacked detailed amenities data, which was central to our analysis. Given these limitations, we concluded that constructing a reliable New York City dataset with both price and amenities information was not feasible within the scope of this project. As a result, we adjusted our research design and shifted our focus to comparing Jersey City and Boston, where more complete and consistent data were available, allowing for a more robust analysis.

Another significant challenge involved processing the amenities data. The original data contained numerous inconsistencies, including spelling variations, duplicate entries, brand names, and extraneous details such as availability times. Substantial effort was required to clean and standardize these entries into meaningful categories, for example, grouping different types of televisions or gaming devices while still preserving important distinctions. Determining the appropriate level of filtering also proved challenging. Applying a high threshold removed too many amenities and limited the analysis, while a lower threshold risked including less meaningful features. Additionally, ensuring consistency between the Boston and Jersey City datasets required further alignment, as the cleaned data were initially stored in different formats. During the development of the Shiny application, we encountered challenges in ensuring that the app consistently used the cleaned and processed data rather than the raw inputs, which occasionally led to unexpected results. Managing how amenities were stored, whether as lists or text, also required careful handling to ensure that filtering and selection features functioned correctly. Overall, the process required multiple rounds of data cleaning, validation, and debugging to produce a reliable dataset and a smooth, user-friendly application.

Figure 1(Airbnb website)

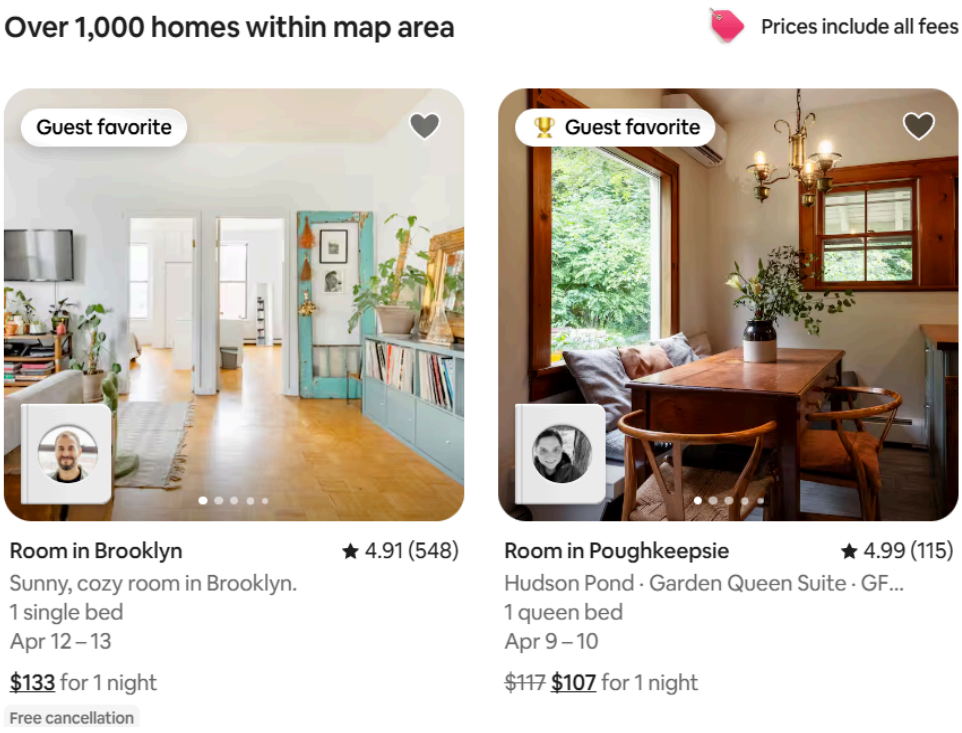


Figure 2(Kaggle dataset)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	id	name	host_id	host_name	neighbourl	neighbourl	latitude	longitude	room_type	price	minimum_	number_of	last_review	reviews_pe	calculated	availability_365	
2	2539	Clean & qu	2787	John	Brooklyn	Kensingtor	40.64749	-73.9724	Private roo	149	1	9	########	0.21	6	365	
3	2595	Skylit Midtc	2845	Jennifer	Manhattan	Midtown	40.75362	-73.9838	Entire hom	225	1	45	########	0.38	2	355	
4	3647	THE VILLA(	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.9419	Private roo	150	3	0			1	365	
5	3831	Cozy Entire	4869	LisaRoxani	Brooklyn	Clinton Hill	40.68514	-73.9598	Entire hom	89	1	270	7/5/2019	4.64	1	194	
6	5022	Entire Apt:	7192	Laura	Manhattan	East Harle	40.79851	-73.944	Entire hom	80	10	9	########	0.1	1	0	
7	5099	Large Cozy	7322	Chris	Manhattan	Murray Hill	40.74767	-73.975	Entire hom	200	3	74	########	0.59	1	129	
8	5121	BlissArtsSp	7356	Garon	Brooklyn	Bedford-St	40.68688	-73.956	Private roo	60	45	49	########	0.4	1	0	
9	5178	Large Furn	8967	Shunichi	Manhattan	Hell's Kitch	40.76489	-73.9849	Private roo	79	2	430	########	3.47	1	220	
10	5203	Cozy Clear	7490	MaryEllen	Manhattan	Upper Wes	40.80178	-73.9672	Private roo	79	2	118	########	0.99	1	0	
11	5238	Cute & Coz	7549	Ben	Manhattan	Chinatown	40.71344	-73.9904	Entire hom	150	1	160	6/9/2019	1.33	4	188	
12	5295	Beautiful 1	7702	Lena	Manhattan	Upper Wes	40.80316	-73.9655	Entire hom	135	5	53	########	0.43	1	6	

Data Cleaning: **Prices**

The Airbnb datasets for Boston and Jersey City were carefully cleaned to ensure consistency and usability for analysis. Listings with missing or empty price values were removed, as price is the primary variable of interest. Price data, originally stored as strings with symbols such as dollar signs and commas, was converted to a numeric format for accurate calculations. Extremely high-priced listings, often reflecting long-term rentals with minimum stays of 120 nights or more, were filtered out, keeping only listings under $1,000 to focus on reasonable, short-term accommodations. Finally, prices were adjusted to include applicable short-term rental taxes as of September 2025, which were approximately 15.25% for Boston (state, local, and convention center taxes) and 9.625% for Jersey City (state sales and local occupancy taxes). This ensured that comparisons between cities reflected the full cost to guests, providing a meaningful basis for analyzing how amenities and other factors influence Airbnb pricing.

(Figure 3)

Raw Prices Before Cleaning:

price <chr>
$125.00
$129.00
$168.00
$140.00
$166.00
$202.00

(Figure 4)

Prices After Cleaning, Adjusting for Tax:

price_with_tax <dbl>
144.0625
148.6725
193.6200
161.3500
191.3150
232.8050

Data Cleaning: **Amenities**

The amenities variable required extensive preprocessing because it was originally stored as a single text string for each listing, often containing multiple items and extraneous text, rather than structured, separate variables. Before data cleaning, the amenities dataset contained 2,879 unique entries for Boston and 1,514 for Jersey City **(Fig 5,6,7)**. These figures included inconsistencies such as misspellings, varied descriptions, brand-specific names, and multiple ways of referring to the same amenity. After cleaning, these entries were standardized and grouped into broader, consistent categories of unique variables.

The first step involved breaking these strings into individual items and removing obviously irrelevant listings that contained no usable information. This allowed the analysis to focus only on listings with meaningful amenities data.

Once separated, the text was standardized and normalized to ensure consistency across listings. Variations in capitalization, spacing, punctuation, and character encoding were corrected so that different forms of the same amenity, for example, “TV with Netflix,” “tv with netflix,” and “TV with Netflix ”would be treated as equivalent. Irrelevant fragments, such as descriptive phrases or scheduling information (“available upon request,” “always at the listing”), were removed to reduce noise and ensure that only functional amenities were retained.

The next stage involved grouping similar items into broader, interpretable categories. Television-related items were consolidated into a single “TV” category, internet features into “internet,” and similar items like refrigerators, ovens, stoves, and coffee machines were standardized under unified labels. Lifestyle and functional groupings were also applied, combining entertainment features into categories such as “game console” and “sound system,” leisure and outdoor amenities into “patio/balcony,” “gym/workout equipment,” or “pool,” and safety features into clear labels like smoke alarms or first aid kits. This step ensured that the analysis reflected the function of each amenity rather than the exact wording chosen by hosts.

After standardizing the data, duplicate entries within individual listings were removed to ensure that no single amenity was counted more than once per property. A frequency-based filtering approach was then applied to retain only the most commonly occurring amenities. However, certain rare yet meaningful amenities, such as game consoles and pianos, were manually preserved to maintain their relevance. A final manual review was conducted to address any remaining inconsistencies or overly specific entries, merging similar items under consistent labels. Following this cleaning process, a total of 106 unique amenities were identified for Boston and 101 for New Jersey **(Figures 8 and 9)**. To visualize the results, bar plots were created showing the top 10 most common amenities for Airbnb listings across all locations in both Boston and New Jersey.

(Figure 5,6,7) Amenities data after cleaning:

```
====================================
BOSTON RAW AMENITY STRINGS (NO CLEANING)
====================================
Total raw unique amenity strings: 2879

[[1]]
 [1] "c(c(c(c(c(c(heating"          "dishwasher"
 [3] "smoke alarm"                  "iron"
 [5] "carbon monoxide alarm"        "cooking basics"
 [7] "hair dryer"                   "essentials"
 [9] "refrigerator"                 "patio/balcony"
[11] "shampoo"                      "hangers"
[13] "internet"                     "washer"
[15] "extra pillows and blankets"   "bed linens"
[17] "oven"                         "dryer"
[19] "kitchen"                      "microwave"
[21] "stove"                        "parking"
[23] "tv"                           "coffee maker"
[25] "air conditioning/fan"         "dishes and silverware"
[27] "hot water)))))"

[[2]]
 [1] "c(c(c(c(c(c(heating"          "smoke alarm"
 [3] "private entrance"             "iron"
 [5] "carbon monoxide alarm"        "first aid kit"
 [7] "hair dryer"                   "essentials"
 [9] "refrigerator"                 "shampoo"
[11] "self check-in"                "hangers"
[13] "internet"                     "long term stays allowed"
```

```
====================================
JERSEY - RAW AMENITY STRINGS (NO PROCESSING)
====================================
Total raw unique amenity strings: 1514

[[1]]
 [1] "refrigerator"             "essentials"
 [3] "washer"                   "kitchen"
 [5] "first aid kit"            "heating"
 [7] "hangers"                  "parking"
 [9] "hair dryer"               "microwave"
[11] "high chair"               "stove"
[13] "dishes and silverware"    "dedicated workspace"
[15] "internet"                 "game console"
[17] "air conditioning/fan"     "smoke alarm"
[19] "carbon monoxide alarm"    "long term stays allowed"
[21] "host service"             "crib"
[23] "children's dinnerware"    "tv"
[25] "fireplace"                "dryer"
[27] "fire extinguisher"        "shampoo"
[29] "luggage dropoff allowed"  "breakfast"
[31] "single level home"        "hot water"
[33] "oven"                     "dishwasher"
[35] "books and toys"           "coffee maker"
[37] "gym/workout equipment"    "cooking basics"
[39] "iron"                     "changing table"

[[2]]
 [1] "refrigerator"              "essentials"
```

amenities
<chr>

["Heating", "Dishwasher", "Smoke alarm", "Iron", "Carbon monoxide alarm", "Cooking basics", "Hair dryer", "Essentials", "Refrigerator", "Patio or...
["Heating", "Smoke alarm", "Private entrance", "Iron", "Carbon monoxide alarm", "First aid kit", "Hair dryer", "Essentials", "Refrigerator", "Shamp...
["Heating", "Dishwasher", "Smoke alarm", "Iron", "Carbon monoxide alarm", "Cooking basics", "Hair dryer", "Essentials", "Refrigerator", "Shampo...
["Heating", "Smoke alarm", "Elevator", "Iron", "Carbon monoxide alarm", "Cooking basics", "Hair dryer", "Essentials", "Refrigerator", "Patio or bal...
["Heating", "Smoke alarm", "Iron", "Carbon monoxide alarm", "Hair dryer", "Essentials", "Pets allowed", "Shampoo", "Hangers", "Wifi", "Long ter...
["Toaster", "Heating", "Dishwasher", "Smoke alarm", "Iron", "Carbon monoxide alarm", "Baking sheet", "Cooking basics", "Hair dryer", "Essential...

(Figure 8,9)
Amenities data after cleaning:

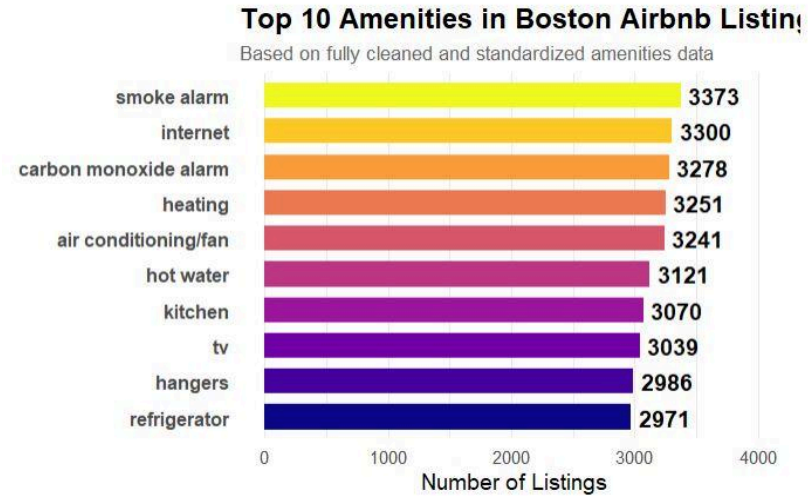
```
Total unique cleaned amenities: 106

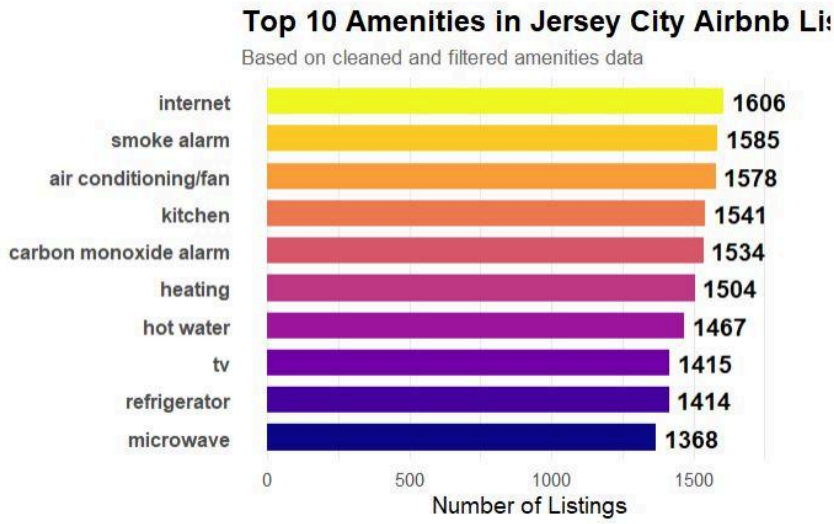
  [1] "air conditioning/fan"       "baby bath"
  [3] "baby monitor"               "babysitter recommendations"
  [5] "backyard"                   "baking sheet"
  [7] "bathtub/bidet"              "bbq grill"
  [9] "bbq utensils"               "beach access"
 [11] "bed linens"                 "bikes"
 [13] "blender"                    "boat slip"
 [15] "body soap"                  "books and toys"
 [17] "bowling alley"              "bread maker"
 [19] "breakfast"                  "carbon monoxide alarm"
 [21] "changing table"             "child safety features"
 [23] "children's dinnerware"      "cleaning products"
 [25] "climbing wall"              "clothing storage"
 [27] "coffee maker"               "conditioner"
 [29] "cooking basics"             "crib"
 [31] "dedicated workspace"        "dining table"
 [33] "dishes and silverware"      "dishwasher"
 [35] "dryer"                      "drying rack for clothing"
 [37] "dvd player"                 "elevator"
 [39] "essentials"                 "ev charger"
 [41] "extra pillows and blankets" "fire extinguisher"
 [43] "fire pit"                   "fireplace"
 [45] "first aid kit"              "freezer"
 [47] "game console"               "game table"
 [49] "games"                      "gym/workout equipment"
 [51] "hair dryer"                 "hangers"
 [53] "heating"                    "high chair"
```

```
Final number of Jersey amenities: 101

  [1] "air conditioning/fan"       "baby bath"
  [3] "baby monitor"               "babysitter recommendations"
  [5] "backyard"                   "backyard fully fenced"
  [7] "baking sheet"               "bathtub/bidet"
  [9] "bbq grill"                  "bbq utensils"
 [11] "bed linens"                 "blender"
 [13] "boat slip"                  "body soap"
 [15] "books and toys"             "breakfast"
 [17] "carbon monoxide alarm"      "changing table"
 [19] "children's dinnerware"      "cleaning during stay"
 [21] "cleaning products"          "clothing storage"
 [23] "coffee maker"               "conditioner"
 [25] "cooking basics"             "crib"
 [27] "dedicated workspace"        "dining table"
 [29] "dishes and silverware"      "dishwasher"
 [31] "dryer"                      "drying rack for clothing"
 [33] "dvd player"                 "elevator"
 [35] "essentials"                 "ev charger"
 [37] "extra pillows and blankets" "fire extinguisher"
 [39] "fire pit"                   "fireplace"
 [41] "first aid kit"              "freezer"
 [43] "game console"               "game table"
```

Figure(10,11)
Top 10 amenities in Boston and Jersey after cleaning





Simple Visualizations of the Data:

The summary statistics and box plot show a clear difference between the Airbnb markets of Boston and Jersey City, with Boston maintaining a higher central tendency across all primary statistics. Specifically, Boston's mean price of $236.02 and median of $202.00 is significantly more expensive than Jersey City's mean of $169.87 and median of $139.00, suggesting that, as a whole, Jersey City serves as a more budget-friendly baseline for students seeking temporary housing. This difference is shown by the interquartile ranges in the box plot, as the middle 50% of Boston's listings are priced notably higher than those in Jersey City. Furthermore, the higher standard deviation in Boston (*SD*=$162.52) compared to Jersey City (*SD*=$124.15) indicates greater price volatility and a wider variety of listing types in the Boston market. Boston's higher variability suggests a more segmented market with both budget and luxury tiers, whereas Jersey City appears more uniform and price-constrained. While both cities show a dense cluster of outliers toward our $1,000 threshold, the overall distribution confirms that while both markets scale into the luxury tier, the "typical" stay in Jersey City remains consistently more affordable than its Boston counterpart.

Figure 12: Side-by-Side Boxplot of Total Price

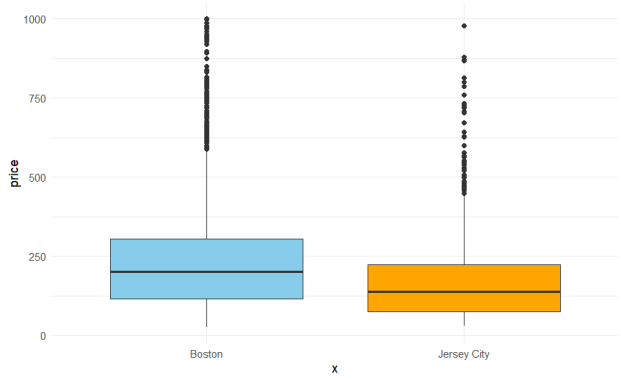


Figure 13: Summary Statistics

Statistic	Boston	Jersey_City
Mean	236.02	169.87
Median	202.00	139.00
SD	162.52	124.15
Min	26.00	30.00
Max	1000.00	978.00
Q1	117.50	76.00
Q3	306.00	225.00

By adjusting the total listing cost to the price per guest metric, we can more accurately evaluate the economic value offered by each city, controlling for the variations in capacity observed in the previous histograms. The data shows a contrast in affordability as Boston's median price per guest stands at $66.17, nearly 71% higher than Jersey City's median of $38.67. This discrepancy directly addresses our research question by highlighting that even when listing size is standardized, Boston remains a significantly more expensive market. Furthermore, the interquartile range (IQR) for Boston ($45.00 to $100.50) is much broader than Jersey City's ($29.50 to $52.36), suggesting that while Jersey City offers a more predictable and compact pricing structure, Boston's prices fluctuate more based on other characteristics. These visualizations provide the baseline to investigate the "amenity premium" central to our study. The presence of numerous high-value outliers in both cities, reaching as high as $500.00 per guest in Boston, indicates that certain listings are commanding prices far above the market average.

Figure 14: Side-by-Side Boxplot of Price Per Guest

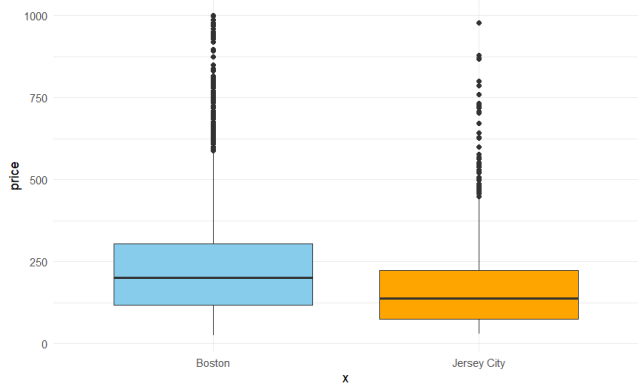


Figure 15: Summary Statistics

Statistic	Boston	Jersey_City
Mean	79.31	44.68
Median	66.17	38.67
SD	47.47	25.39
Min	9.29	8.75
Max	500.00	360.00
Q1	45.00	29.50
Q3	100.50	52.36

To contextualize the findings, it is important to distinguish between the "Neighborhoods" used in the Boston dataset and the "Wards" used in Jersey City. While a neighborhood like Dorchester or Back Bay is typically defined by shared cultural identity, historical boundaries, or commercial hubs, a Ward is a formal political subdivision used for municipal governance and city council representation. In Jersey City, Wards (A through F) are used more frequently than neighborhoods in official datasets and analyses because they are legally defined, unambiguous, and widely recognized for administrative purposes. The use of Wards as a primary geographic filter reflects the city's administrative structure, where each Ward is represented by a specific council member. For our research, this means the Jersey City data is aggregated into six large, legally defined districts, whereas the Boston data is broken down into smaller, more residential communities.

To understand the geographic distribution of our data, we visualized the volume of listings across various neighborhoods, which reveals distinct market structures in each city. In Jersey City, listings are relatively well-distributed across its political Wards, with Ward F and Ward D leading the count, suggesting a broad availability of short-term rentals throughout the city's residential zones. In contrast, Boston's market is highly concentrated in specific neighborhoods like Dorchester, Back Bay, and Downtown, while areas like the Leather District offer almost no listings. This geographic density is important to our research question, as neighborhood desirability often dictates the baseline price before amenities are even considered. For a student traveler, these charts suggest that while Jersey City provides a more uniform spread of options, Boston's supply is heavily skewed toward specific hubs, likely driving the higher price-per-guest and standard deviation observed in our previous metrics. By identifying where these listings are clustered, we can better assess whether the pricing premiums we find are a result of location or the specific luxury amenities offered within those high-demand neighborhoods.

Figure 16: Bar Graph of Airbnb Distribution in Boston

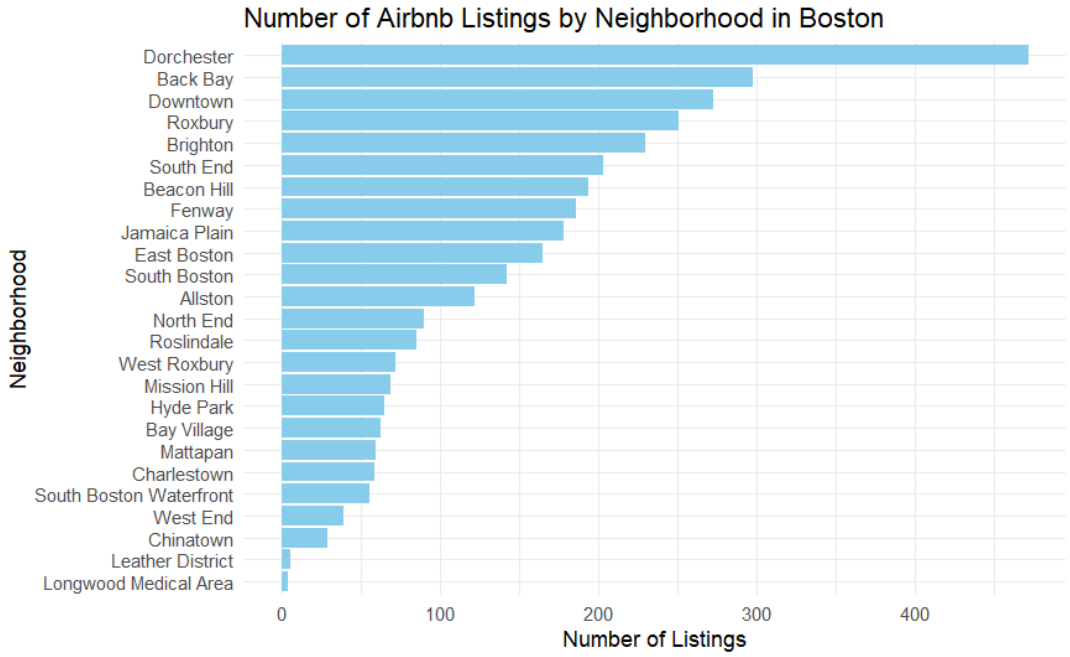
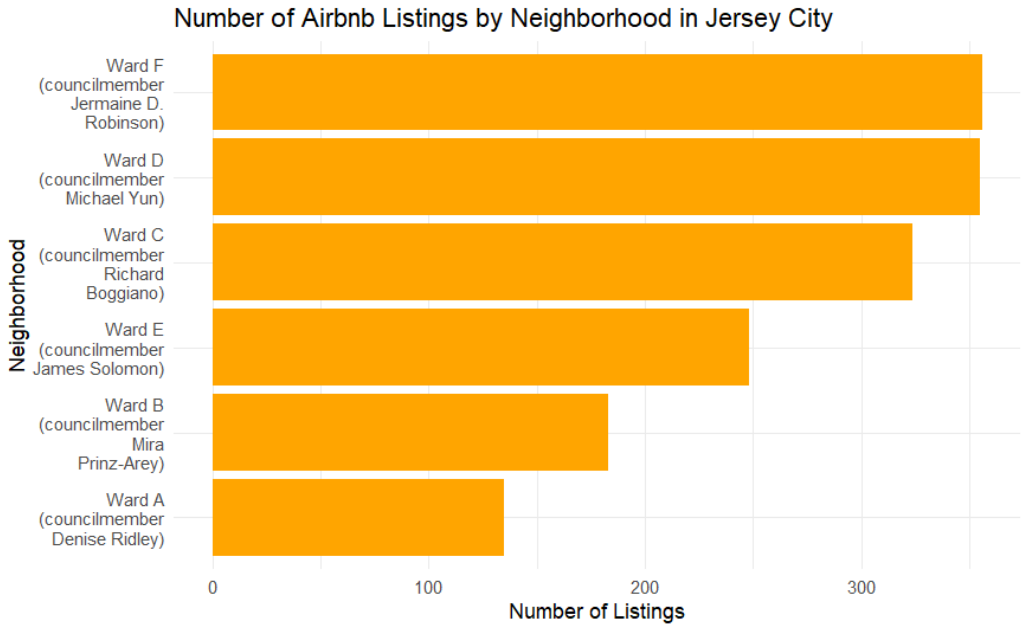


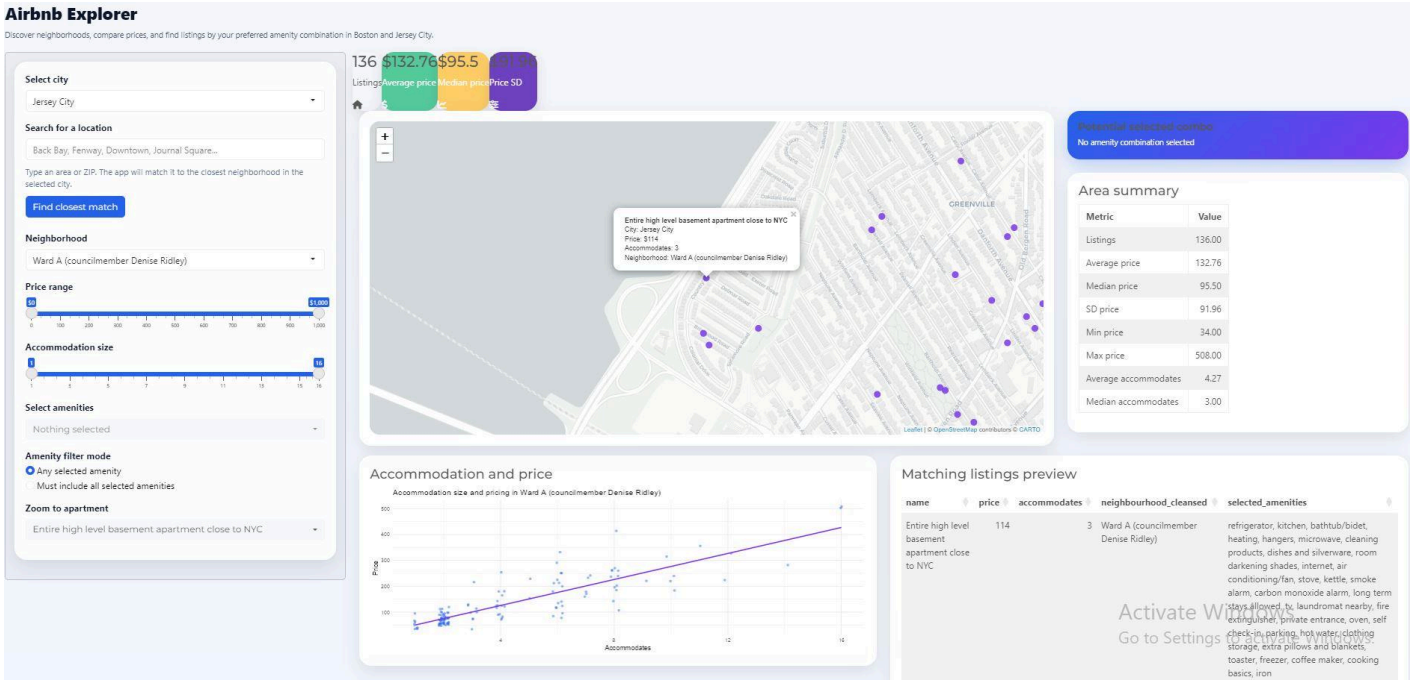
Figure 17: Bar Graph of Airbnb Distribution in Jersey City



Shiny interactive app

The Shiny application was developed to provide an interactive and user-friendly platform for exploring Airbnb listings in Boston and Jersey City based on student preferences. Following the cleaning and standardization of the amenities data, the objective was to transform the dataset into a practical decision-support tool that helps users understand how factors such as location, accommodation size, and amenities influence pricing. The Explorer allows users to select a city, search for specific neighborhoods, and filter listings based on price range and accommodation capacity. A key feature is the ability to select individual amenities or combinations of amenities, enabling users to analyze how specific features such as internet access, parking, or gaming consoles impact both listing availability and pricing. The “Potential Selected Combo” panel provides a clear summary of the selected amenities, making it easy to interpret the applied filters. In addition, the interface includes an interactive map that visualizes listing locations and allows users to zoom in on individual properties. Summary statistics, including average price, median price, and price variability, are dynamically updated based on user selections. A listings table further complements this by presenting detailed information on properties that match the selected criteria. Overall, the Explorer bridges the gap between raw data analysis and practical usability. It enables users to explore patterns, test hypotheses, and gain insights into how amenities and location influence Airbnb pricing in a clear, visual, and interactive manner.

Figure 18: Shiny app display



Hypothesis Testing:

Question: Overall, is Boston more expensive than Jersey City?

A Welch two-sample t-test was conducted to determine whether there is a significant difference in Airbnb price per guest between Boston and Jersey City. The results show a highly significant difference between the two cities ($t = 33.582$, $df = 4930.1$, $p < 2.2 \times 10^{-16}$), indicating strong evidence against the null hypothesis of equal means. The average price per guest in Boston ($79.31) is substantially higher than in Jersey City ($44.68). The 95% confidence interval for the difference in means ranges from $32.60 to $36.65, suggesting that Boston listings are, on average, approximately $33–$37 more expensive per guest than those in Jersey City. Overall, this provides strong statistical evidence that Airbnb listings in Boston are significantly more expensive per guest than those in Jersey City.

Figure 19: T-Test result of Boston vs Jersey City Price per Guest

```
        Welch Two Sample t-test

data:  price_per_guest by city
t = 33.582, df = 4930.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Boston and group Jersey City is not equal to 0
95 percent confidence interval:
 32.60388 36.64659
sample estimates:
     mean in group Boston mean in group Jersey City
                 79.30643                  44.68120
```

Question: Do amenities affect the price per guest?

To examine how specific amenities affect Airbnb pricing in Jersey City, we first created binary indicator variables for ten common amenities: tv, dishes and silverware, gym/workout equipment, parking, cooking basics, pets allowed, heating, patio balcony, hot water, and air conditioning/fan. Using these indicators, we wrote a loop in R that performs a Welch two-sample t-test for each amenity, comparing the mean price per guest for listings with the amenity with listings without the amenity. The results from this loop were stored in a data frame that includes the t-statistic, degrees of freedom, p-value, and the mean price per guest for listings with and without the amenity.

To examine the impact of specific amenities on Airbnb pricing in Boston, we created indicator variables for ten key amenities: TV, dishes and silverware, gym or workout equipment, parking, cooking basics, pets allowed, heating, patio or balcony, hot water, and air conditioning or fan. These amenities were selected because they reflect common preferences among students seeking short-term stays during vacations, school breaks, conferences, and similar occasions. Using these indicators, we conducted Welch two-sample t-tests for each amenity to compare the mean price per guest between listings that offered the amenity versus those that did not. The results revealed clear patterns in how amenities influence pricing. For example, listings with gyms, pets allowed, or air conditioning/fans were significantly more expensive per guest, suggesting a positive "amenity premium" for features that enhance comfort or convenience. Conversely, what was surprising was that listings with TVs, dishes, and silverware, parking, patio/balcony, or hot water were less expensive, indicating that these amenities may be standard expectations that do not increase price substantially. Amenities such as cooking basics and heating showed no significant difference in price per guest, suggesting that these features do not meaningfully alter market valuation. While the t-test results provide useful initial insight, they do not control for other factors such as city, listing size, or additional amenities. As a result, these differences should not be interpreted as causal effects. To address this limitation, we incorporate these variables into a multiple regression framework in the following section. Overall, these analyses provide insight into how specific listing features contribute to variation in Airbnb pricing, highlighting which amenities drive higher prices and which are more neutral in effect.

Figure 20: T-Test result of Boston amenities with vs without an amenity

	amenity <chr>	t_stat <dbl>	df <dbl>	p_value <dbl>	mean_with <dbl>	mean_without <dbl>
t	tv	-13.0334640	669.2552	9.140549e-35	81.90659	58.80987
t1	dishes and silverware	15.0970328	1089.2976	6.648835e-47	70.86614	104.45990
t2	gym/workout equipment	-18.3386717	811.6524	4.099243e-63	112.14014	72.36891
t3	parking	14.1262781	2535.1418	1.134932e-43	70.67132	94.06638
t4	cooking basics	-0.2639341	811.1907	7.918977e-01	79.43020	78.77160
t5	pets allowed	-12.4625109	2172.8495	1.751119e-34	93.17882	71.47299
t6	heating	-0.6590143	200.4164	5.106430e-01	79.42697	77.13005
t7	patio/balcony	11.4225458	1804.0310	3.173277e-29	65.16358	83.87626
t8	hot water	3.2091926	329.1332	1.462149e-03	78.12143	91.33075
t9	air conditioning/fan	-4.7550890	213.6648	3.645240e-06	80.04662	66.09751

The t-test results reveal that nearly all analyzed amenities significantly influence listing prices in Boston, except cooking basics and heating, which both have p-values well above the 0.05 threshold. The most dramatic price premium is associated with gym/workout equipment, where listings featuring a gym average $112.14 per guest compared to only $72.37 for those without. Other amenities like TV, pets allowed, and air conditioning also show statistically significant correlations with higher prices. Conversely, some amenities like dishes and silverware and parking actually show significantly lower mean prices when present, suggesting these may be more common in budget-friendly or high-occupancy listings. Overall, the data suggests that while basic utilities are price-neutral, premium additions like fitness facilities allow hosts to command a substantial financial premium.

Figure 21: T-Test result of Jersey City amenities with vs without an amenity

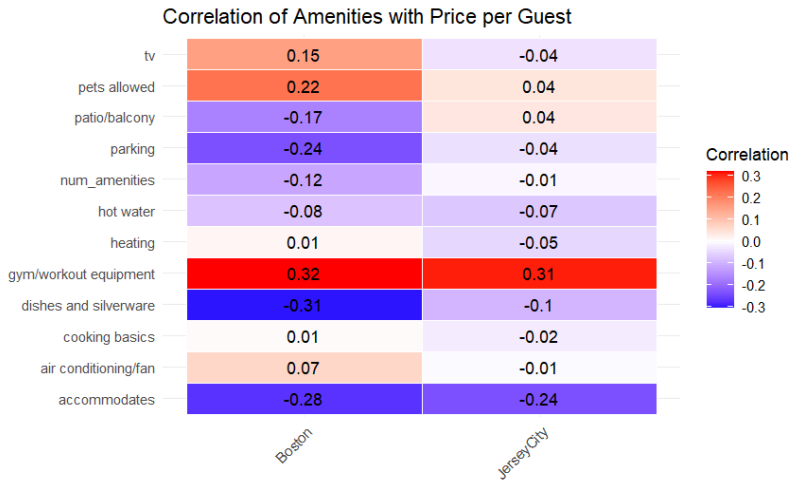
	amenity <chr>	t_stat <dbl>	df <dbl>	p_value <dbl>	mean_with <dbl>	mean_without <dbl>
t	tv	1.2720849	248.83094	2.045305e-01	44.33146	47.06287
t1	dishes and silverware	2.9661742	307.47516	3.251639e-03	43.55053	50.35581
t2	gym/workout equipment	-9.1655206	216.98508	3.853691e-17	65.51842	41.77437
t3	parking	1.3750417	889.25010	1.694648e-01	44.02973	46.00887
t4	cooking basics	0.7497669	479.81782	4.537626e-01	44.39265	45.70505
t5	pets allowed	-1.5988541	578.56886	1.103990e-01	46.54968	44.16226
t6	heating	1.5786953	123.75954	1.169586e-01	44.33853	49.15091
t7	patio/balcony	-1.5970849	609.28816	1.107653e-01	46.47673	44.19164
t8	hot water	1.8886834	156.97940	6.077881e-02	44.11439	50.28762
t9	air conditioning/fan	0.2677211	45.38231	7.901271e-01	44.65219	45.70795

The t-test results for Jersey City also indicate that most amenities do not have a statistically significant impact on Airbnb pricing, as their p-values exceed the 0.05 threshold. Amenities such as TV, parking, cooking basics, pets allowed, heating, patio or balcony, hot water, and air conditioning or fan show no meaningful difference in average price between listings that include them and those that do not. However, two amenities stand out as having a significant influence on pricing: gym or workout equipment and dishes and silverware. Gym access, in particular, has a very strong effect, with listings offering this amenity averaging approximately $65.52 per guest compared to $41.77 for those without, indicating a substantial price premium. Dishes and silverware also show a statistically significant impact, though the difference in average price is smaller (about $43.55 versus $50.36). Overall, these findings suggest that while most common amenities do not meaningfully affect Airbnb pricing in Jersey City, higher-value or lifestyle-oriented features such as access to a gym can significantly increase listing prices.

Correlations:

Correlation analyses were conducted to explore the relationships between Airbnb listing prices per guest and various listing characteristics in both Boston and Jersey City. First, the correlation between price_per_guest and the number of guests accommodated (accommodates) showed weak negative relationships in Boston (-0.28) and Jersey City (-0.24), indicating that larger accommodations tend to have lower per-guest prices. The total number of amenities (num_amenities) was also examined, showing similarly weak correlations (-0.12 in Boston; -0.01 in Jersey), suggesting that more amenities do not strongly drive per-guest pricing. Individual amenities were further analyzed for their association with price_per_guest. For Boston, amenities such as tv, dishes and silverware, gym/workout equipment, parking, cooking basics, pets allowed, heating, patio/balcony, hot water, and air conditioning/fan all showed small to moderate correlations ranging roughly from -0.31 to 0.32, with some positive (e.g., gym/workout equipment, 0.32) and some negative (e.g., dishes and silverware, -0.31). Because amenity variables are binary indicators, these correlations represent differences in average price between listings with and without each amenity. Similar trends were observed in Jersey City, where correlations were mostly low to moderate in magnitude, indicating that no single amenity strongly predicts per-guest price. Still, certain features, such as gym/workout equipment (0.31), tend to be associated with higher prices, while others, like hot water (-0.07) or heating (-0.05), show weaker or negative associations. Overall, these results suggest that a combination of factors influences Airbnb per-guest pricing, but no single feature overwhelmingly determines pricing.

Figure 22: Correlation table of different amenities



Assumptions:

To model Airbnb price per guest, we applied a logarithmic transformation to the dependent variable (price_per_guest) to address its right-skewed distribution and stabilize variance across listings. This transformation helps satisfy the regression assumption of normally distributed residuals and reduces the influence of extreme values, allowing coefficient estimates to be interpreted in relative percentage terms. We then fitted a multiple linear regression model that included the city, number of guests accommodated (accommodates), selected individual amenities (e.g., TV, parking, gym/workout equipment, patio/balcony, pets allowed), and neighborhood fixed effects (neighbourhood_cleansed) to examine both within- and cross-city pricing differences.

Key assumptions of linear regression were carefully evaluated to ensure model validity. Residual versus fitted value plots were used to assess linearity and homoscedasticity, confirming that residuals were roughly centered around zero and exhibited constant variance. QQ plots indicated that residuals approximated a normal distribution, supporting the appropriateness of the log transformation. Multicollinearity among predictors was assessed using Variance Inflation Factors (VIF), all of which were well below the commonly used threshold of 5, indicating reliable and interpretable coefficient estimates. Finally, the model assumes independence of residuals across listings, consistent with the sampling design. Together, these steps ensure that Model 5 provides a robust and interpretable framework for understanding how listing characteristics, location, and city influence Airbnb prices per guest.

Figure 23: Residual and QQplot of Model before log transformation

Before Log Transformation:

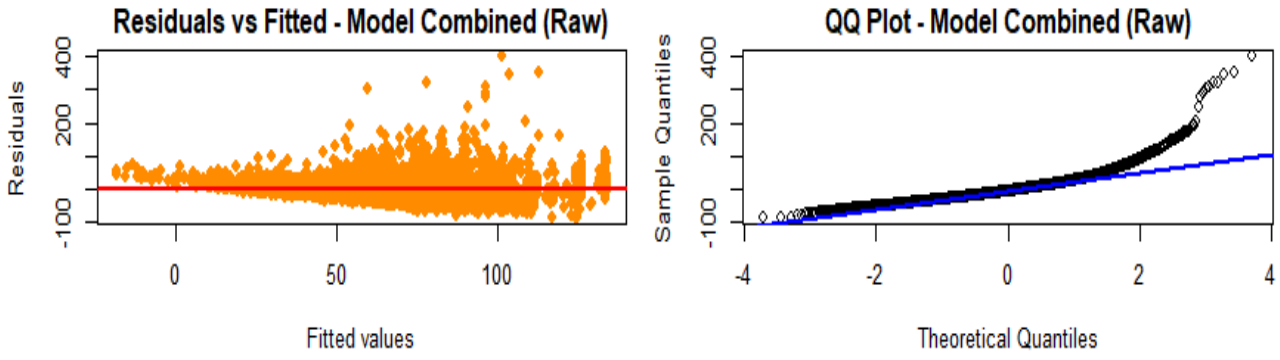


Figure 24: Residual and QQplot of Model after log transformation

After Log Transformation:

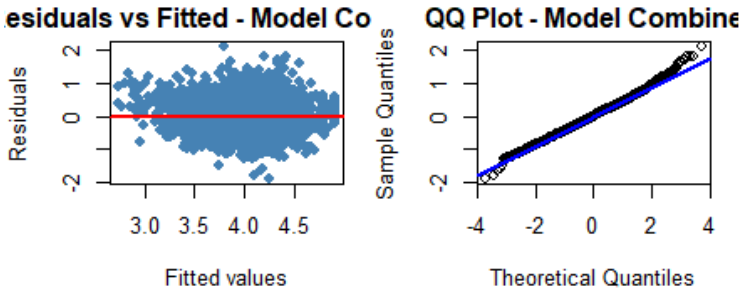


Figure 25: VIF table of the different amenities

VIF of Model 5:

```
                    city             accommodates           num_amenities
tv
                1.106944                 1.224640                 2.841559
1.185583
`gym/workout equipment`                  parking          `pets allowed`   `air
conditioning/fan`
                1.253002                 1.233498                 1.212727
1.101754
`dishes and silverware`         `cooking basics`                  heating
`patio/balcony`
                1.856369                 1.610569                 1.675560
1.266076
             `hot water`
                1.778375
```

Multiple Linear Regression:

The results of the multiple linear regression model indicate several important factors influencing Airbnb price per guest across Boston and Jersey City. The model explains approximately 39% of the variation in log-transformed price per guest (Adjusted $R^2 = 0.394$), and the overall F-test ($F = 253.5$, $p < 0.001$) confirms that the included predictors jointly have a significant effect on pricing. Holding other variables constant, listings in Jersey City are associated with log-prices approximately 0.432 units lower than those in Boston ($\beta = -0.432$, $p < 2e\text{-}16$), highlighting substantial cross-city differences. The number of guests accommodated is negatively associated with price per guest ($\beta = -0.068$, $p < 2e\text{-}16$), indicating that larger listings tend to have lower per-guest costs. Conversely, the total number of amenities has a small but significant positive effect ($\beta = 0.004$, $p = 2.82e\text{-}05$), suggesting that more feature-rich listings command slightly higher prices.

Figure 26: Coefficient Table

Model Results:

```
Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              4.1606446  0.0403741 103.052  < 2e-16 ***
cityJersey City         -0.4320736  0.0143525 -30.104  < 2e-16 ***
accommodates            -0.0676984  0.0026594 -25.456  < 2e-16 ***
num_amenities            0.0038948  0.0009292   4.192 2.82e-05 ***
tv                       0.1906730  0.0215056   8.866  < 2e-16 ***
`gym/workout equipment`  0.2906225  0.0195290  14.882  < 2e-16 ***
parking                 -0.1241261  0.0147502  -8.415  < 2e-16 ***
`pets allowed`           0.1057904  0.0150784   7.016 2.59e-12 ***
`air conditioning/fan`   0.0899631  0.0322419   2.790 0.005287 **
`dishes and silverware` -0.2277330  0.0207838 -10.957  < 2e-16 ***
`cooking basics`         0.0771494  0.0202515   3.810 0.000141 ***
heating                  0.0671918  0.0350931   1.915 0.055592 .
`patio/balcony`         -0.0166776  0.0168884  -0.988 0.323435
`hot water`             -0.0633965  0.0295821  -2.143 0.032156 *
```

Individual amenities also exhibit varied impacts on pricing. Features such as a TV ($\beta = 0.191$, $p < 2e\text{-}16$), gym/workout equipment ($\beta = 0.291$, $p < 2e\text{-}16$), and pet allowances ($\beta = 0.106$, $p = 2.59e\text{-}12$) are associated with higher per-guest prices, while parking ($\beta = -0.124$, $p < 2e\text{-}16$) and dishes and silverware ($\beta = -0.228$, $p < 2e\text{-}16$) are linked to lower prices. Other amenities, including air conditioning/fan ($\beta = 0.090$, $p = 0.005$) and cooking basics ($\beta = 0.077$, $p = 0.000141$), show smaller positive effects, whereas heating, patio/balcony, and hot water have negligible or marginally significant associations ($p = 0.056$, 0.323, and 0.032, respectively). Overall, these results suggest that both location and specific listing characteristics contribute meaningfully to Airbnb pricing, with certain amenities substantially increasing per-guest costs and larger accommodations reducing them, while Boston consistently commands a premium relative to Jersey City.

Figure 27: Scatterplot of Price per Night fitted with a linear model

Scatterplot of the Predicted vs Actual Price Per Night for Airbnb in Jersey and Boston

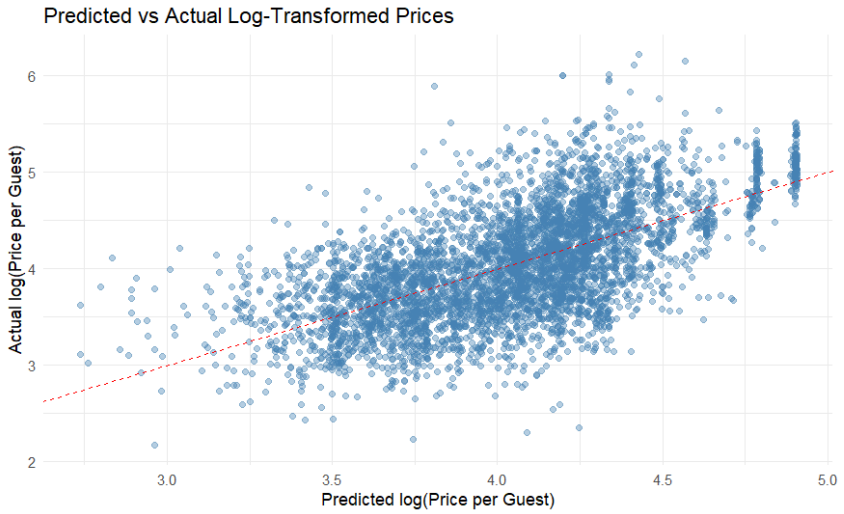
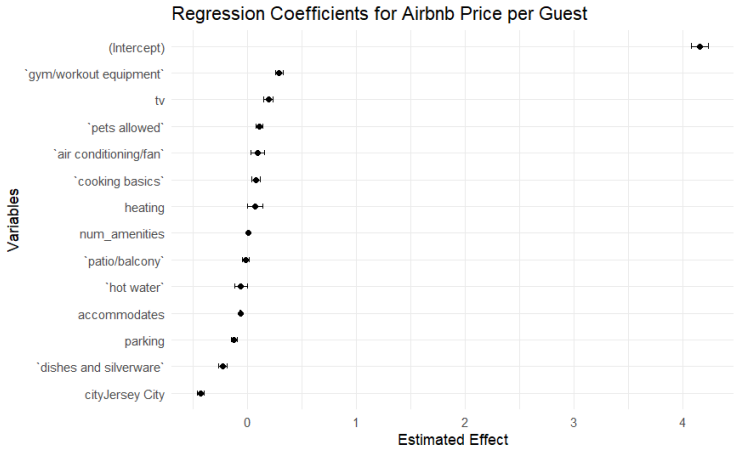


Figure 28: Regression coefficient plot



Interactions:

We fitted a multiple linear regression model to examine how Airbnb price per guest (log-transformed) varies with city, listing characteristics, and amenities, including interactions between city and specific amenities. In the city × amenity interaction model, several predictors had significant effects. For example, the main effect of city was significant ($\beta = -0.380$, $p < 0.001$), indicating that listings in Jersey City are generally lower-priced than those in Boston. Among amenities, gym/workout equipment ($\beta = 0.266$, $p < 0.001$), TV ($\beta = 0.281$, $p < 0.001$), and pets allowed ($\beta = 0.110$, $p < 0.001$) were associated with higher prices per guest, while dishes and silverware showed a negative effect ($\beta = -0.227$, $p < 0.001$). Notably, several city-amenity interactions were significant, such as cityJersey City: tv ($\beta = -0.250$, $p < 0.001$) and cityJersey City: patio/balcony ($\beta = 0.241$, $p < 0.001$), suggesting that the value of these amenities differs between Boston and Jersey City; for instance, a patio or balcony increases price more in Jersey City compared to Boston, whereas a TV increases price more in Boston. The model explained approximately 41.6% of the variation in log(price_per_guest) (Adjusted $R^2 = 0.413$).

We then added an interaction between accommodates and the total number of amenities to explore whether the incremental value of amenities depends on listing size. This more complex model showed that the accommodates × num_amenities interaction was positive and highly significant (β = 0.0013, $p < 0.001$), indicating that listings with more guests gain a larger per-guest price premium from additional amenities. Overall, ANOVA comparisons supported these model extensions: including city × amenity interactions significantly improved model fit over the baseline model ($F = 16.27$, $p < 0.001$), and further including accommodates × num_amenities also improved fit ($F = 38.15$, $p < 0.001$).

These results indicate that Airbnb pricing is context-dependent: the effect of specific amenities on price per guest varies by city, and the effect of total amenities is moderated by the size of the listing. This highlights the importance of both location and listing characteristics in explaining variation in Airbnb pricing, demonstrating that pricing strategies should consider not only the presence of amenities but also city-specific and size-specific interactions.

Logistic Regression:

A logistic regression was conducted to examine the factors associated with a listing being classified as under 100 per night among properties accommodating two people or fewer. The dependent variable, under_100, was binary, with 1 indicating a price below 100 and 0 otherwise. The model included predictors such as city, number of guests accommodated, number of amenities, and specific amenities like TV, gym equipment, parking, and pets allowed.

The results indicate that being located in Jersey City (cityJersey City) significantly increases the likelihood of a listing being under $100 (estimate = 2.167, p < 2e-16$). Several amenities were also significant predictors: listings with a TV (estimate = -1.666), gym/workout equipment (estimate = -1.418), pets allowed (estimate = -1.085), or air conditioning/fan (estimate = -0.612) were less likely to be under 100, whereas the presence of parking (estimate = 0.939), dishes and silverware (estimate = 1.190), and a patio/balcony (estimate = 0.488) increased the odds. Some variables, such as heating, hot water, and the total number of amenities, were not statistically significant, suggesting they do not strongly affect the probability of a listing being affordable in this subset.

Figure 29: Logistic Regression Coefficient Table

```
Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)            1.971089   0.382985   5.147 2.65e-07 ***
cityJersey City        2.167094   0.200071  10.832  < 2e-16 ***
accommodates           0.298883   0.156396   1.911 0.055996 .
num_amenities         -0.005020   0.008808  -0.570 0.568711
tv                    -1.666387   0.220904  -7.543 4.58e-14 ***
`gym/workout equipment` -1.418447   0.188889  -7.509 5.94e-14 ***
parking                0.938862   0.124352   7.550 4.35e-14 ***
`pets allowed`        -1.084663   0.132271  -8.200 2.40e-16 ***
`air conditioning/fan` -0.611806   0.264617  -2.312 0.020775 *
`dishes and silverware` 1.190155   0.173389   6.864 6.69e-12 ***
`cooking basics`      -0.578830   0.175072  -3.306 0.000946 ***
heating               -0.179705   0.274475  -0.655 0.512646
`patio/balcony`        0.488411   0.173391   2.817 0.004850 **
`hot water`           -0.042885   0.228510  -0.188 0.851132
```

The difference in deviance between the null model and the full model is 971.56 with 13 degrees of freedom, which is highly significant ($p < 2.2e-16$). This indicates that adding these predictors significantly improves the model's ability to explain the likelihood of a listing being under 100, compared to a model with no predictors. In other words, the included features provide meaningful information about price classification, and the large reduction in deviance demonstrates that the model fits the data substantially better than an intercept-only model.

Figure 30: ANOVA table results

```
Analysis of Deviance Table

Model 1: under_100 ~ 1
Model 2: under_100 ~ city + accommodates + num_amenities + tv + `gym/workout equipment` +
    parking + `pets allowed` + `air conditioning/fan` + `dishes and silverware` +
    `cooking basics` + heating + `patio/balcony` + `hot water`
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      2462     2954.9
2      2449     1983.3 13   971.56 < 2.2e-16 ***
```

Conclusion:

This study set out to examine how Airbnb prices per guest differ between Jersey City and Boston and to what extent these differences can be explained by listing characteristics such as amenities, neighborhood location, and the number of guests accommodated. Across all analyses, the results consistently demonstrate that Boston is significantly more expensive than Jersey City, even after adjusting for listing size and amenities. Both the Welch two-sample t-test and multiple regression results confirm a substantial price premium in Boston, with listings costing roughly 40–70% more per guest on average. This indicates that broader market conditions, such as demand, tourism intensity, and urban structure, play a major role in determining pricing differences between the two cities.

At the same time, listing characteristics meaningfully contribute to price variation within each city. The regression models show that the number of guests accommodated is a key driver, with larger listings having lower per-guest costs, reinforcing the idea that smaller listings are relatively more expensive. Amenities also play an important but nuanced role. Certain features, such as gym/workout equipment, patios/balconies, and pet allowances, are associated with higher prices, suggesting they function as "premium" amenities that increase perceived value. In contrast, more standard features, such as dishes and silverware or parking, are associated with lower or negligible price effects, indicating that they may be baseline expectations rather than differentiating factors in the market.

Importantly, the interaction models reveal that the value of amenities is not constant across cities. Instead, the effect of specific amenities depends on location, with some features (such as TVs or patios/balconies) contributing differently to price in Boston versus Jersey City. Additionally, the positive interaction between accommodations and the number of amenities suggests that larger listings benefit more from additional amenities, highlighting that pricing is influenced by a combination of listing size and feature offerings rather than any single factor in isolation. These findings emphasize that Airbnb pricing is highly context-dependent, shaped by both city-level dynamics and listing-specific characteristics.

From a practical perspective, these results are especially relevant for college students seeking affordable short-term housing. The logistic regression analysis focusing on listings under $100 per night shows that affordability is significantly less likely in Boston, and that many desirable amenities reduce the likelihood of finding budget-friendly options. However, features such as parking or basic kitchen amenities may increase the chances of finding lower-cost listings. Combined with the finding that smaller listings tend to have higher per-guest prices, this suggests that students aiming to minimize costs may benefit from considering slightly larger accommodations or prioritizing essential over luxury amenities.

Overall, this study demonstrates that while amenities and listing features do influence Airbnb pricing, city-level differences remain the dominant factor, with Boston consistently commanding higher prices than Jersey City. By jointly analyzing location, amenities, and listing size, this research provides a more comprehensive understanding of how short-term rental markets operate across urban environments. These insights not only answer the research question but also offer practical guidance for both hosts optimizing pricing strategies and travelers seeking cost-effective housing options.

Strengths and Weaknesses:

One of the key strengths of this study is the comprehensive and structured approach used to analyze Airbnb pricing across two cities. The data underwent extensive cleaning and standardization, particularly for price and amenities, which improved the reliability and comparability of the results. Converting prices to a per-guest basis allowed for a more meaningful comparison across listings of different sizes, directly addressing a common limitation in housing price analysis. Additionally, the use of multiple statistical methods, including t-tests, multiple linear regression, interaction models, and logistic regression, provided a well-rounded understanding of pricing dynamics. The inclusion of interaction terms was particularly valuable, as it captured the context-dependent nature of amenities and demonstrated that their effects vary across cities and listing sizes. Together, these approaches strengthen the validity of the findings and provide both statistical rigor and practical insight.

However, the study also has several limitations. First, the dataset is limited to Boston and Jersey City, which restricts the generalizability of the results to other cities with different housing markets or tourism patterns. Second, while amenities were carefully cleaned and categorized, there may still be measurement error due to variation in how hosts report amenities or differences in quality that are not captured by binary indicators. Third, the models explain only about 40% of the variation in price per guest, indicating that important factors such as property type, host reputation, reviews, proximity to attractions, and seasonal demand were not included in the analysis. Additionally, the cross-sectional nature of the data limits the ability to make causal claims, as observed relationships reflect associations rather than direct effects. Finally, the logistic regression threshold (e.g., under $100) is somewhat arbitrary and may not fully capture affordability across different users or time periods. Despite these limitations, the study provides a strong foundation for understanding Airbnb pricing and highlights areas for future research.

Contributions:

Winson and Daniel both equally contributed to this report/project. Winson mainly worked on cleaning the prices, data visualizations, hypothesis testing, and modelling(multiple linear regression, logistic regression). Daniel mainly worked on cleaning the amenities, clustering, and working on the Shiny app. We both worked on writing and organizing the final report as well as the presentation.